

PPG INDUSTRY— TALK 1



Introduction

PPG Industries, Inc. is an American supplier of paints, coatings, and specialty materials based in Pittsburgh, Pennsylvania. It operates in more than 70 countries globally including Malaysia. PPG Industry has long taken the initiative to offer industrial training opportunities to students. For several years, it has consistently taken Computer Science students from UTM, especially, allowing them to gain practical experience in a real-world industry setting and apply the knowledge they have learned at university. The main goal of this talk is to help students understand what it's really like to work with data in the industry and how it differs from what we usually do in class. While university assignments often use small and simple datasets, in the real world, data can be massive, messy, and way more complex. This session will share lessons learned from actual industry experience, including how to handle large datasets, optimize performance, design smooth workflows, and make sure data is accurate and reliable. You'll also learn how to take the concepts you've studied in class and apply them to solve real-world problems confidently. By the end of the talk, students will have a clearer idea of what to expect in a professional data analytics environment and how to be prepared to contribute effectively from day one.

Executive Summary

Data Analytics by Derrick Tan Hong Shi, Data Analytics Manager

The PPG's Data Analytics team primarily facilitates data-driven decision-making by analyzing raw data to identify insights, forecast trends, and detect operational inefficiencies. Such applications is used to systems including the Global Marine Planning system for shipping risk prediction and the Complexity Analysis Tool for supply chain and sourcing risk management. It is structured into four primary areas: Master Data, ERP Analytics, Data Analytics, and ML Ops. Required skills of an individual would include proficiency in SQL and Python, data pipeline tools such as Azure Data Factory and Databricks, Power BI for visualization, DevOps practices, and database design.

Introduction to Global Infrastructure by Vijaykumar Dhandapani, Cloud Architecture Lead

To maintain competitiveness in a rapidly evolving digital landscape, PPG aims to move away from mainframe systems and embrace modern cloud-based architectures—including IaaS, PaaS, SaaS, and the growing use of iPaaS for integration. This shift also involves adopting AI-driven IT operations (AIOps) to automate processes and improve predictive maintenance, alongside a robust zero-trust security model to mitigate emerging cyber threats. PPG's IT infrastructure demonstrates this transition in practice. Operating on a clear Plan-Build-Run model where strategy is defined, services are built, and 24/7 operations are maintained. PPG utilizes a multi-cloud ecosystem such as Azure, AWS, VMware, Oracle, and Alibaba Cloud. Such operation requires expertise in cloud platforms, Infrastructure as Code such as Terraform, automation tools like Python and Azure DevOps, networking, and database administration.

SAP Introduction by Patrick Thong, SAP Operations DTR Lead

This session introduces SAP as a leading Enterprise Resource Planning (ERP) platform that integrates business functions into a unified system. At PPG, SAP has been in use for 30 years and now runs on the S/4HANA platform. The system supports PPG's core business processes—such as procurement, production, sales, and finance—through dedicated modules, ensuring a single source of truth across operations. A key ongoing initiative is the "Connect PPG" project, which aims to consolidate over 30 separate ERP systems globally into one unified SAP environment within the next few years. Much of this work is supported by PPG's Shared Services Centre in Shah Alam, Malaysia, which functions as a global IT hub, connecting teams that collaborate on worldwide projects.

Content of Talk

Data Analytics

- Data analytics turns raw data into insights for better decisions.
- It improves efficiency by finding problems and bottlenecks.
- It predicts trends to help businesses plan ahead.
- SQL and database design are essential core skills.

Introduction to Global Infrastructure

- PPG uses a Plan-Build-Run model to manage global IT operations.
- The company uses a cloud-first approach with Azure, AWS, VMware, Oracle, and Alibaba Cloud.
- Infrastructure covers servers, networks, security, and cloud platforms.

SAP Introduction

- SAP is an ERP system that integrates all major business functions.
- PPG is transitioning from Oracle to SAP for better data consistency.
- SAP supports real-time business reporting and analytics.



**Qistina Batrisyia
binti Noor Mohd
Azlan
A25CS340**

REFLECTION:

The PPG talk motivated me to improve my data engineering skills and showed how data roles shape career opportunities. It inspired me to pursue an internship at PPG to grow my expertise and contribute meaningfully.



**Mursyidah binti
Jahidi
A25CS0286**

REFLECTION:

Overall, the talk made me more aware of the challenges and expectations in the data analytics field and motivated me to strengthen my skills in data modeling, visualization, and pipeline development.



**Nor Ain Fahira
binti Muhamad
Fariq
A25CS0295**

REFLECTION:

The talk showed me that teamwork and technical knowledge go hand-in-hand in IT, and it also reminded me that soft skills like communication, problem-solving, and adaptability are just as important to develop throughout my degree.